Mathematical Formulation of Special Grain Boundaries and Associated Boundary Conditions in the Mesoscopic Tensorial Model (MTM)

Coincidence Site Lattice (CSL) Rationalization, Seam Overlap Resolution, Analytical Pre-Opening, Energy-Monotone Delaunay Surgery, and Two-Stage Relaxation–Loading Boundary Algebra

MTM Lattice-Triangulation Documentation

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Abstract

This document presents the complete mathematical formulation and algorithmic architecture for constructing special Coincidence Site Lattice (CSL) grain boundaries and enforcing their associated two-stage boundary conditions (**Relaxation Step BCs** vs. **Loading Step BCs & Indenter**) in the 2D Mesoscopic Tensorial Model (MTM). We derive the coprime Diophantine generators (m, n) for square (C_4) and triangular (C_6) Bravais lattices, the parity halving rule for the coincidence index $\Sigma(m, n)$, the structural repeat period $L_{\mathrm{CSL}}(m, n)$, the interplanar layer spacing $d_{mn} = a_0^2/L_{\mathrm{CSL}}$, and the closed-form analytical normal pre-opening gap $\delta_\perp(m, n)$ that prevents initial cross-boundary triangles from starting inside the repulsive core of the Zanzotto/Ericksen potential. We then formulate the 5-state independent (x, y) boundary condition algebra (`node_bc_role` $\in \{0, 1, 2, 3, 4\}$), the orthogonal subspace projection operator $\mathbf{P}_{\mathrm{BC}}$, the proof that the 3-Roller + 1-Free `GB-Frame Roller` configuration eliminates all three 2D rigid-body zero modes while guaranteeing $\sigma^{\mathrm{eq}}_{\perp\perp} = 0$, and the reference handover operator $(\mathbf{X}_0 \leftarrow \mathbf{x}_{\mathrm{eq}})$ that eliminates boundary snap upon transitioning from Stage 1 relaxation to Stage 2 mechanical loading or nanoindentation.

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1 2D Crystallographic Kinematics and Rotated Grain Lattices

1.1 Bravais Lattice Bases and Point-Group Symmetry

Let $a_0 > 0$ denote the fundamental equilibrium lattice parameter ($a_0 = 1$ in normalized lattice units). A 2D Bravais lattice $\mathcal{L}_0 \subset \mathbb{R}^2$ is generated by integer linear combinations of two primitive basis vectors $\mathbf{a}_1, \mathbf{a}_2 \in \mathbb{R}^2$:

$$\mathcal{L}_0 = \left\{ \mathbf{X}_{i,j} = i\,\mathbf{a}_1 + j\,\mathbf{a}_2 \;\middle|\; (i,j) \in \mathbb{Z}^2 \right\} = \mathbf{A}_0\,\mathbb{Z}^2, \qquad \mathbf{A}_0 = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 \end{bmatrix}. \tag{1}$$

Two fundamental 2D crystallographic symmetry classes are implemented:

1. **Square Lattice (C_4/D_4 symmetry, `square`)**:

$$\mathbf{a}_1 = a_0 \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad \mathbf{a}_2 = a_0 \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad \mathbf{A}_0^{\mathrm{sq}} = a_0 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad A_{\mathrm{cell}}^{\mathrm{sq}} = \det \mathbf{A}_0^{\mathrm{sq}} = a_0^2. \tag{2}$$

   By D_4 dihedral symmetry, all physically distinct symmetric tilt misorientations lie in $\Delta\theta \in [0°, 90°]$, with mirror duality across $\Delta\theta = 45°$.

2. **Triangular / Hexagonal Lattice (C_6/D_6 symmetry, `triangular`)**:

$$\mathbf{a}_1 = a_0 \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad \mathbf{a}_2 = a_0 \begin{bmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{bmatrix}, \qquad \mathbf{A}_0^{\mathrm{tri}} = a_0 \begin{bmatrix} 1 & \frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} \end{bmatrix}, \qquad A_{\mathrm{cell}}^{\mathrm{tri}} = \det \mathbf{A}_0^{\mathrm{tri}} = \frac{\sqrt{3}}{2} a_0^2. \tag{3}$$

   By D_6 dihedral symmetry, all physically distinct symmetric tilt misorientations lie in $\Delta\theta \in [0°, 60°]$, with duality across $\Delta\theta = 30°$.

For a grain g with crystallographic orientation $\theta_g \in \mathbb{R}$ and origin anchor $\mathbf{p}_g \in \mathbb{R}^2$, the rotated lattice basis is $\mathbf{A}_g = \mathbf{R}(\theta_g)\mathbf{A}_0$, where

$$\mathbf{R}(\theta_g) = \begin{bmatrix} \cos\theta_g & -\sin\theta_g \\ \sin\theta_g & \cos\theta_g \end{bmatrix} \in \mathrm{SO}(2), \qquad \mathcal{L}_g(\theta_g, \mathbf{p}_g) = \left\{ \mathbf{p}_g + \mathbf{R}(\theta_g)\mathbf{A}_0 \begin{bmatrix} i \\ j \end{bmatrix} \;\middle|\; (i,j) \in \mathbb{Z}^2 \right\}. \tag{4}$$

1.2 Bicrystal Grain Boundary Plane Geometry

In a macroscopic domain $\Omega = [x_{\min}, x_{\min} + L_x] \times [y_{\min}, y_{\min} + L_y]$, a planar bicrystal interface Γ_{GB} is specified by normalized pivot coordinates $(c_x, c_y) \in [0,1]^2$ and normal inclination angle ϕ_{GB} (`gb_angle_deg`):

$$\mathbf{p}_{\mathrm{GB}} = \begin{bmatrix} x_{\min} + c_x L_x \\ y_{\min} + c_y L_y \end{bmatrix}, \qquad \hat{\mathbf{n}}_{\mathrm{GB}}(\phi_{\mathrm{GB}}) = \begin{bmatrix} \cos\phi_{\mathrm{GB}} \\ \sin\phi_{\mathrm{GB}} \end{bmatrix}, \qquad \hat{\mathbf{t}}_{\mathrm{GB}}(\phi_{\mathrm{GB}}) = \begin{bmatrix} -\sin\phi_{\mathrm{GB}} \\ \cos\phi_{\mathrm{GB}} \end{bmatrix}. \tag{5}$$

The signed normal distance $s(\mathbf{x})$ and tangential coordinate $\tau(\mathbf{x})$ of any point $\mathbf{x} \in \Omega$ are

$$s(\mathbf{x}) = (\mathbf{x} - \mathbf{p}_{\mathrm{GB}}) \cdot \hat{\mathbf{n}}_{\mathrm{GB}}, \qquad \tau(\mathbf{x}) = (\mathbf{x} - \mathbf{p}_{\mathrm{GB}}) \cdot \hat{\mathbf{t}}_{\mathrm{GB}}. \tag{6}$$

For a symmetric tilt grain boundary (STGB) with half-tilt angle θ and total misorientation $\Delta\theta = 2\theta$, the two half-crystals have orientations

$$\theta_1 = \phi_{\mathrm{GB}} - \theta = \phi_{\mathrm{GB}} - \frac{\Delta\theta}{2} \quad (s \le 0), \qquad \theta_2 = \phi_{\mathrm{GB}} + \theta = \phi_{\mathrm{GB}} + \frac{\Delta\theta}{2} \quad (s \ge 0). \tag{7}$$

2 Coincidence Site Lattice (CSL) Theory and General (m, n) Rationalizer

2.1 Diophantine Coincidence Condition on a 2D Square Lattice

Consider two square lattices $\mathcal{L}_1 = \mathbf{R}(-\theta)\mathbb{Z}^2 a_0$ and $\mathcal{L}_2 = \mathbf{R}(+\theta)\mathbb{Z}^2 a_0$ meeting along a vertical boundary plane ($\phi_{\mathrm{GB}} = 0°$, $\hat{\mathbf{t}}_{\mathrm{GB}} = \hat{\mathbf{e}}_y$). A lattice site $\mathbf{p} = a_0(n\hat{\mathbf{e}}_x + m\hat{\mathbf{e}}_y) \in \mathbb{Z}^2 a_0$ lies strictly on the vertical boundary axis after rotation by $+\theta$ if and only if its x-component vanishes:

$$\mathbf{R}(\theta)\begin{bmatrix} na_0 \\ ma_0 \end{bmatrix} = a_0\begin{bmatrix} n\cos\theta - m\sin\theta \\ n\sin\theta + m\cos\theta \end{bmatrix} = \begin{bmatrix} 0 \\ L_{\mathrm{CSL}} \end{bmatrix}. \tag{8}$$

This yields the coprime integer generator law ($m, n \in \mathbb{Z}_{>0}$, $\gcd(m, n) = 1$, $1 \le n \le m$):

$$\tan\theta_{mn} = \frac{n}{m}, \qquad \theta_{mn} = \arctan\left(\frac{n}{m}\right), \qquad \Delta\theta_{mn} = 2\arctan\left(\frac{n}{m}\right). \tag{9}$$

The relative rotation matrix between the two grains is then a rational orthogonal matrix:

$$\mathbf{R}(\Delta\theta_{mn}) = \frac{1}{m^2+n^2}\begin{bmatrix} m^2-n^2 & -2mn \\ 2mn & m^2-n^2 \end{bmatrix} \in \mathrm{SO}(2, \mathbb{Q}). \tag{10}$$

2.2 The Coincidence Index $\Sigma(m, n)$ and Parity Halving Rule

The reciprocal density of shared coincidence sites on the square lattice is given by

$$\Sigma_{\mathrm{sq}}(m, n) = \frac{m^2+n^2}{\gcd(m^2+n^2,\, 2)}. \tag{11}$$

Parity Halving and 90° Duality of Square-Lattice CSL Boundaries.
Because $\gcd(m, n) = 1$, m and n cannot both be even:

- **Opposite parity ($m + n$ odd $\implies$ $m^2 + n^2$ odd)**: $\gcd(m^2 + n^2, 2) = 1$, so $\Sigma(m, n) = m^2 + n^2$. For $(m, n) = (2, 1)$, $\Sigma = 5$ at $\Delta\theta = 2\arctan(1/2) \approx 53.130°$ ($\Sigma 5\,(210)$).
- **Both odd (m, n odd $\implies$ $m^2 + n^2$ even)**: $(m \pm n)$ are both even, so the centered vector $\frac{1}{2}((m + n)\mathbf{a}_1 + (m - n)\mathbf{a}_2) \in \mathbb{Z}^2 a_0$ halves the CSL unit cell area, yielding $\Sigma(m, n) = (m^2 + n^2)/2$. For $(m, n) = (3, 1)$, $\Sigma = 10/2 = 5$ at $\Delta\theta = 2\arctan(1/3) \approx 36.870°$ ($\Sigma 5\,(310)$).

Under the dual mapping $(m', n') = (m + n, m - n)$, $\tan\theta_{m',n'} = \tan(45° - \theta_{mn})$, so $\Delta\theta_{m',n'} = 90° - \Delta\theta_{mn}$ and $\Sigma(m', n') = \Sigma(m, n)$.

On the 2D triangular lattice ($\mathbf{a}_1 = a_0(1, 0)^{\mathsf{T}}$, $\mathbf{a}_2 = a_0(1/2, \sqrt{3}/2)^{\mathsf{T}}$), the corresponding coprime generator law for $\gcd(m, n) = 1$ ($1 \le n < m$) is:

$$\tan\theta^{\mathrm{tri}}_{mn} = \frac{\sqrt{3}\,n}{2m+n}, \qquad \Delta\theta^{\mathrm{tri}}_{mn} = 2\arctan\left(\frac{\sqrt{3}\,n}{2m+n}\right), \qquad \Sigma_{\mathrm{tri}}(m, n) = \frac{m^2+mn+n^2}{\gcd(m^2+mn+n^2,\, 3)}. \tag{12}$$

2.3 Structural Period L_{CSL}, Layer Spacing d_{mn}, and Commensurate Sizing

The structural repeat period along Γ_{GB} and the interplanar spacing of atomic layers parallel to Γ_{GB} are:

$$L_{\mathrm{CSL}}(m,n) = a_0\sqrt{m^2+n^2}, \qquad d_{mn} = \frac{a_0^2}{L_{\mathrm{CSL}}(m,n)} = \frac{a_0}{\sqrt{m^2+n^2}}. \tag{13}$$

To ensure an integer number $K_{\mathrm{CSL}} \geq 2$ of full structural units along the boundary length, the specimen grid size N^{CSL} around a user target N_{target} is quantized as

$$K_{\mathrm{CSL}} = \max\!\Big(2,\ \mathrm{round}\Big(\frac{N_{\mathrm{target}}\,a_0}{L_{\mathrm{CSL}}(m,n)}\Big)\Big), \qquad N^{\mathrm{CSL}}(m,n) = \mathrm{round}\Big(K_{\mathrm{CSL}}\sqrt{m^2+n^2}\Big). \tag{14}$$

Table 1: Catalog of Special 2D Square-Lattice CSL Symmetric Tilt Grain Boundaries ($\Sigma \leq 65$).

(m,n)	m^2+n^2	Σ	Plane (hkl)	θ_{mn}	$\Delta\theta_{mn} = 2\theta_{mn}$	L_{CSL}/a_0	d_{mn}/a_0	$\delta_\perp(m,n)/a_0$
$(1,0)$	1	Σ1	(100)	0.000°	0.000°	1.0000	1.0000	0.0000
$(9,1)$	82	Σ41a	(910)	6.340°	12.680°	9.0554	0.1104	0.2774
$(7,1)$	50	Σ25a	(710)	8.130°	16.260°	7.0711	0.1414	0.3743
$(6,1)$	37	Σ37a	(610)	9.462°	18.925°	6.0828	0.1644	0.2824
$(5,1)$	26	Σ13a	(510)	11.310°	22.620°	5.0990	0.1961	0.1555
$(4,1)$	17	Σ17a	(410)	14.036°	28.072°	4.1231	0.2425	0.4549
$(7,2)$	53	Σ53a	(720)	15.945°	31.891°	7.2801	0.1374	0.3906
$(3,1)$	10	Σ5a	(310)	18.435°	36.870°	3.1623	0.3162	0.3075
$(5,2)$	29	Σ29a	(520)	21.801°	43.603°	5.3852	0.1857	0.1972
$(2,1)$	5	Σ5b	(210)	26.565°	53.130°	2.2361	0.4472	0.0456
$(7,4)$	65	Σ65a	(740)	29.745°	59.490°	8.0623	0.1240	0.1958
$(5,3)$	34	Σ17b	(530)	30.964°	61.928°	5.8310	0.1715	0.2540
$(3,2)$	13	Σ13b	(320)	33.690°	67.380°	3.6056	0.2774	0.3853
$(4,3)$	25	Σ25b	(430)	36.870°	73.740°	5.0000	0.2000	0.1400
$(5,4)$	41	Σ41b	(540)	38.660°	77.320°	6.4031	0.1562	0.3153
$(1,1)$	2	Σ1	(110)	45.000°	90.000°	1.4142	0.7071	0.0000

3 Microscopic Degrees of Freedom, Seam Overlap Resolution, and Pre-Opening

3.1 Coincidence Capture Band and Rigid Half-Crystal Translations

Both half-crystals are anchored at $\mathbf{p}_{\mathrm{GB}}$ using a symmetric coincidence capture band $\epsilon_{\mathrm{band}} = 0.02\,a_0$ so floating-point roundoff ($\sim 10^{-15}\,a_0$) never drops shared coincidence sites on $s(\mathbf{X}) = 0$:

$$\text{Grain 0:}\quad \mathbf{X}^{(0)}_{i,j} = \mathbf{p}_{\mathrm{GB}} + i\,\mathbf{R}(\theta_1)\mathbf{a}_1 + j\,\mathbf{R}(\theta_1)\mathbf{a}_2, \qquad s(\mathbf{X}^{(0)}_{i,j}) \leq +\epsilon_{\mathrm{band}}, \tag{15}$$

$$\text{Grain 1:}\quad \mathbf{X}^{(1)}_{i,j} = \tilde{\mathbf{X}}^{(1)}_{i,j} + \delta_{\parallel} a_0\,\hat{\mathbf{t}}_{\mathrm{GB}} + \delta_\perp a_0\,\hat{\mathbf{n}}_{\mathrm{GB}}, \qquad s(\tilde{\mathbf{X}}^{(1)}_{i,j}) \geq -\epsilon_{\mathrm{band}}, \tag{16}$$

where $\delta_{\parallel}$ (`gb_shift_tangent_a0`) and $\delta_\perp$ (`gb_normal_gap_a0`) are the microscopic rigid-body translations parallel and normal to Γ_{GB}.

3.2 Distance-Sorted Spatial-Hash Seam Overlap Filter

All cross-grain pairs ($p \in G_0$, $q \in G_1$) with Euclidean separation $r_{pq} = \|\mathbf{X}_p - \mathbf{X}_q\|_2 < d_{\min}a_0$ (default $d_{\min} = 0.45$) are extracted via a uniform spatial hash of cell size $h = d_{\min}a_0$ and sorted in ascending order of r_{pq}. Under the default `MergeMidpoint` policy, each non-removed pair (p,q) is replaced by its symmetric midpoint

$$\mathbf{X}_{\mathrm{mid}} = \frac{1}{2}\,(\mathbf{X}_p + \mathbf{X}_q)\,, \tag{17}$$

subject to the secondary crowding guard $\|\mathbf{X}_k - \mathbf{X}_{\text{mid}}\|_2 \geq 0.85\, d_{\min} a_0$ for all surviving third neighbors $k \notin \{p, q\}$. This places shared CSL structural-unit vertices strictly on the mirror symmetry plane of the boundary.

3.3 Derivation of the Analytical Normal Pre-Opening Gap $\delta_\perp(m, n)$

Across a symmetric tilt boundary with layer spacing $d_{mn} = a_0/\sqrt{m^2+n^2}$, mirror-symmetric atomic planes of index $k \in \mathbb{Z}_{>0}$ sit at $s = \pm k\, d_{mn}$ with cross-boundary separation $r_k = 2k\, d_{mn}$. Since all layers with $r_k < d_{\min} a_0$ are fused by the overlap filter, the first unmerged cross-boundary layer pair has index and separation:

$$k_{\text{first}}(m,n) = \max\left(1,\ \left\lceil \frac{d_{\min}\sqrt{m^2+n^2}}{2} \right\rceil\right), \qquad r_{\text{first}}(m,n) = 2\, k_{\text{first}}(m,n)\, d_{mn}. \tag{18}$$

When $r_{\text{first}}(m,n) < 0.94\, a_0$, the initial Delaunay triangles bridging the seam are born under severe normal compression ($\det \mathbf{F}_e \ll 1$). The analytical pre-opening gap

$$\delta_\perp(m,n) = \text{clamp}\left(\frac{0.94\, a_0 - r_{\text{first}}(m,n)}{a_0},\ 0.0,\ 0.42\right) \tag{19}$$

restores the initial cross-seam bond length to $r_{\text{first}} + \delta_\perp a_0 \approx 0.94\, a_0$, placing all interfacial elements inside the quadratic basin of the Zanzotto potential before L-BFGS begins.

4 Constrained Delaunay Triangulation, MTM Energy, and Core Surgery

Each triangle $e \in \mathcal{T}_{\text{act}}$ (filtered by geometric cutouts and maximum chord length $l_{\max} = 2.60\, a_0$) has deformation gradient $\mathbf{F}_e = [\mathbf{x}_{i_1} - \mathbf{x}_{i_0},\, \mathbf{x}_{i_2} - \mathbf{x}_{i_0}]\mathbf{D}_{0,e}^{-1}$ and Cauchy–Green metric $\mathbf{C}_e = \mathbf{F}_e^{\mathsf{T}}\mathbf{F}_e$. Under Gauss–Lagrange reduction $\mathbf{C}_{\text{red},e} = \mathbf{M}_e^{\mathsf{T}}\mathbf{C}_e\mathbf{M}_e \in \mathcal{D}_{\text{red}}$ with $\mathbf{M}_e \in \text{GL}(2,\mathbb{Z})$, the total potential energy and element Piola–Kirchhoff stress are:

$$E_{\text{MTM}}(\{\mathbf{x}_i\},\mathcal{T}_{\text{act}}) = \sum_{e\in\mathcal{T}_{\text{act}}} \frac{A_{0,e}}{A_{\text{cell}}} \Phi_{\text{Zanzotto}}(\mathbf{C}_{\text{red},e}) + E_{\text{indenter}}(\{\mathbf{x}_i\},\alpha), \tag{20}$$

$$\mathbf{P}_e = \frac{2}{A_{\text{cell}}}\, \mathbf{F}_e\, \mathbf{M}_e\, \frac{\partial \Phi_{\text{Zanzotto}}}{\partial \mathbf{C}_{\text{red}}}\, \mathbf{M}_e^{\mathsf{T}}. \tag{21}$$

During energy-driven Delaunay surgery (`perform_remeshing_loop_reduction`), a candidate retriangulation $\mathcal{T}_{\text{cand}} = \text{Delaunay}(\mathbf{x}_{\text{eq}})$ is accepted if and only if $\Delta N_{\text{tris}} > 0$, $\mathcal{T}_{\text{cand}}$ has not been visited in a cycle, and the post-flip relaxed energy strictly decreases:

$$\Delta E_{\text{surgery}} = E(\mathbf{x}_{\text{trial}},\mathcal{T}_{\text{cand}}) - E(\mathbf{x}_{\text{old}},\mathcal{T}_k) < -10^{-6}\,|E(\mathbf{x}_{\text{old}},\mathcal{T}_k)|. \tag{22}$$

5 Mathematics of Associated Boundary Conditions (Relaxation vs. Loading)

5.1 5-State Independent (x, y) Kinematic Classification

Each node $i \in \{1,\ldots,N\}$ accumulates boolean locks $(\ell_{x,i}, \ell_{y,i}) \in \{0,1\}^2$, drive flags $(d_{x,i}, d_{y,i}) \in \{0,1\}^2$, and velocity $\mathbf{v}_i \in \mathbb{R}^2$ across all matching `BoundaryZoneRule` entries, mapping to the 5-state

role `node_bc_role[i]`:

$$\text{role}(i) = \begin{cases} 0 & \text{(Free)}: \quad \ell_{x,i} = 0 \ \wedge \ \ell_{y,i} = 0 \qquad \text{(both } u_{x,i}, u_{y,i} \text{ free)}, \\ 1 & \text{(Fixed)}: \quad \ell_{x,i} = 1 \ \wedge \ \ell_{y,i} = 1 \ \wedge \ \neg(d_{x,i} \vee d_{y,i}) \qquad (u_{x,i} = u_{y,i} = 0), \\ 2 & \text{(Driven)}: \quad \ell_{x,i} = 1 \ \wedge \ \ell_{y,i} = 1 \ \wedge \ (d_{x,i} \vee d_{y,i}) \qquad (\mathbf{u}_i(\alpha) = \alpha \mathbf{v}_i), \\ 3 & \text{(RollerY / X-Locked)}: \quad \ell_{x,i} = 1 \ \wedge \ \ell_{y,i} = 0 \qquad (u_{x,i}(\alpha) = \alpha v_{x,i}, \ u_{y,i} \text{ free}), \\ 4 & \text{(RollerX / Y-Locked)}: \quad \ell_{x,i} = 0 \ \wedge \ \ell_{y,i} = 1 \qquad (u_{x,i} \text{ free}, \ u_{y,i}(\alpha) = \alpha v_{y,i}). \end{cases} \tag{23}$$

At any corner where a `RollerY` edge ($\ell_x = 1$) intersects a `RollerX` edge ($\ell_y = 1$), boolean composition yields $\ell_x = \ell_y = 1$ (`role` = 1, Fixed), pinning the corner automatically.

5.2 Subspace Projection Operator $\mathbf{P}_{\mathbf{BC}}$ and Preconditioned Hessian Restriction

All nodes with role$(i) \in \{0, 3, 4\}$ enter the active solver vector $\mathbf{z} \in \mathbb{R}^{2n_{\text{vars}}}$ with $z_k = x_{i(k)}$ and $z_{n_{\text{vars}}+k} = y_{i(k)}$ ($k = 0, \ldots, n_{\text{vars}} - 1$). The diagonal orthogonal projector $\mathbf{P}_{\text{BC}} = \text{diag}(p_0, \ldots, p_{2n_{\text{vars}}-1})$ zeros the x-component when role $= 3$ and the y-component when role $= 4$. Restricting the sparse CHOLMOD stiffness preconditioner $\mathbf{H}$ to the admissible subspace $\mathcal{V}_{\text{BC}} = \text{im}(\mathbf{P}_{\text{BC}})$ via

$$\tilde{\mathbf{H}} = \mathbf{P}_{\text{BC}} \, \mathbf{H} \, \mathbf{P}_{\text{BC}} + (\mathbf{I} - \mathbf{P}_{\text{BC}}) \tag{24}$$

guarantees that the preconditioned L-BFGS search direction $\mathbf{d} = -\tilde{\mathbf{H}}^{-1}(\mathbf{P}_{\text{BC}} \nabla E)$ satisfies $(\mathbf{I} - \mathbf{P}_{\text{BC}})\mathbf{d} = \mathbf{0}$ identically at every iteration.

5.3 Proof of the Stage 1 Relaxation Step BCs (`GB-Frame Rollers`)

For a vertical bicrystal ($\phi_{\text{GB}} = 0°$), the default **Relaxation Step BCs** assign `Left = RollerY` ($u_x = 0$), `Right = FreeSurface`, and `Top/Bottom = RollerX` ($u_y = 0$):

1. **Rigid-Body Mode Elimination**: Any 2D rigid-body field $\mathbf{u}(x, y) = (c_x - \omega y, \, c_y + \omega x)^{\mathsf{T}}$ satisfying $u_x(x_{\min}, y) = 0$ and $u_y(x, y_{\min}) = 0$ must have $c_x = c_y = \omega = 0$.

2. **Unconstrained Normal Breathing ($\sigma_{xx}^{\text{eq}} = 0$)**: Because the Right Edge is traction-free and the Top/Bottom edges carry zero horizontal shear traction ($P_{xy} = 0$ since u_x is free), global horizontal equilibrium enforces $\langle P_{xx} \rangle_\Omega = 0$. The right half-crystal translates freely along x to accommodate the exact equilibrium GB excess volume $\Delta u_\perp^{\text{eq}}$.

3. **End Wedge Suppression**: Locking $u_y = 0$ on Top and Bottom enforces mirror symmetry across $y = y_{\min}, y_{\max}$ while letting u_x slide freely without Poisson end-pinning.

6 Two-Stage Handover ($\mathbf{X}_0 \leftarrow \mathbf{x}_{\mathbf{eq}}$) and GB Excess Energy $\gamma_{\mathbf{GB}}(\Delta\theta)$

At the transition from Stage 1 (**Relaxation Step BCs**) to Stage 2 (**Loading Step BCs & Indenter**), the solver executes the reference handover operator:

$$\mathbf{X}_{0,i} \longleftarrow \mathbf{x}_{\text{eq},i} \qquad \forall i \in \{1, \ldots, N\}, \tag{25}$$

while keeping the crystallographic element reference matrices $\mathbf{D}_{0,e}$ unchanged. Thus Stage 2 trajectories $\mathbf{x}_i^{\text{Stage 2}}(\alpha) = \mathbf{x}_{\text{eq},i} + \alpha \mathbf{v}_i$ start continuously from $\mathbf{x}_{\text{eq}}$ at $\alpha = 0$ without boundary snap ($\Delta\mathbf{u}_{\text{snap}} = \mathbf{0}$), and Stage 2 nanoindentation re-anchors its contact elevation $y_{\text{top}}^{\text{eq}}$ directly on the relaxed top surface.

Finally, excluding an outer boundary collar of width $w_{\mathrm{collar}} = 4\,a_0$ defines the interior gauge region Ω_{int} and the collar-excluded specific grain boundary excess energy:

$$\gamma_{\mathrm{GB}}^{\mathrm{int}}(\Delta\theta) = \frac{1}{L_{\mathrm{GB}}^{\mathrm{int}}} \sum_{\substack{e \in \mathcal{T}_{\mathrm{eq}} \\ \mathbf{x}_{\mathrm{centroid},e} \in \Omega_{\mathrm{int}}}} \frac{A_{0,e}}{A_{\mathrm{cell}}} \Phi_{\mathrm{Zanzotto}}(\mathbf{C}_{\mathrm{red},e}). \tag{26}$$